

# Steinadler Pro Endurance Testing Summary

## 1. Product Introduction

- **Company Name:** Jiangsu Yinguan Intelligence Technology Co., Ltd.
- **Project Name:** Steinadler Pro
- **Report Date:** September 2, 2025
- **Project Cycle:** April 2023 - July 2024
- **Project Status:** Fully Completed
- **Applicable Scenarios:**
  - *Outdoor Recreation:* the vehicle delivers excellent performance in complex terrains such as deserts and mountains. With its powerful power output and stability, drivers can enjoy adventure experiences and easily traverse obstacles like sand dunes and streams.
  - *Agricultural and Forestry Operations:* on farms and in woodlands, the vehicle enables rapid transportation of materials (e.g., seeds, chemical fertilizers) to improve work efficiency. Forestry personnel can leverage its high ground clearance and four-wheel drive (4WD) system to easily navigate forest trails for patrols or inspections.
  - *Professional Fields:* rescue teams use the vehicle to reach emergency sites quickly, reducing response time; power line maintenance staff can access remote areas conveniently for inspections, thanks to the vehicle's flexibility and passability.
  - *Other potential applicable scenarios:* hunting, fishing or other usage.

## 2. Project Development and Verification

### 2.1 Phase 1: Project Design Verification – 1st Prototype

- **Design & Prototyping Timeline:** product design commenced in April 2023; the first prototype was completed in July 2023.
- **Achievements:** Total 29 quality issues identified in this phase were summarized and rectified, including 22 design issues, 4 incoming material issues, 3 process control issues, 1 tooling fixture issue. All issues have been fully rectified and tracked for implementation.
- **Issue Identification Table:**

| 序号 | 区域 | 零件号            | 日期        | 问题点                                       | 图片                                                                                  | 原因                                                 | 问题归属 | 责任人        | 临时措施     | 状态 | 长期措施                             | 状态   |
|----|----|----------------|-----------|-------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|------|------------|----------|----|----------------------------------|------|
| 1  | 转向 | RJ2302~302-300 | 6.6       | RJ2302-302-311锁扣长螺母无法与RJ2302-302-309连接环固定 |  | 连接环太短                                              | 设计原因 | 刘文浩        | 现场自制螺母固定 | OK | 已更换结构                            | 已实施  |
| 2  | 转向 | RJ2302~302-300 | 6.6       | RI2302-302-008转向轴与RI2302-302-020密封环无法安装   |  | 转向轴外径图纸要求29.966-29.988mm，实测29.96-30mm之间，密封环与图纸尺寸符合 | 来料问题 | 王志翔<br>罗剑兵 | 现场自行打磨   | OK | 加强出货检                            | 已实施  |
| 3  | 转向 | RJ2302~302-300 | 2023.6.19 | 转向轴无法装入                                   |  | 经量测，轴尺寸符合公差。但与轴承实际尺寸存在重合的部分                        | 设计原因 | 卢滢         | 自行打磨轴    | ok | 结构更改，轴承数量变更为1个，装配方式改为压接，需要到料后试装。 | 试装OK |

- **Test Period:** July 3, 2023 – August 3, 2023
- **Test Road Conditions:** Unpaved road conditions (professional racing venue)
- **Test Mileage:** 2,000 km
- **Test Results:** Issues detected during on-road testing are summarized below:
  1. Braking Force: The rear brake feels excessively soft with insufficient feedback force.
  2. Power Performance: Insufficient power output occurred during driving.
  3. Suspension & Chassis: Rear suspension crossbeam was concave; front suspension bearings and steering tie rods could not be fully locked.
  4. Steering System: Steering column displacement; inconsistent left and right steering angles.
- **Follow-up Actions:** all above issues were rectified in this phase. The 2nd prototyping verification was conducted on August 15, 2023.



**1<sup>st</sup> Prototype  
Status**



**Partial Issue  
Photos**



**Road Test Photo**

- **Video:** 1.4 Endurance Testing Phase 1.mp4

## 2.2 Phase 2: Project Design Verification – 2nd Prototype

- **Prototyping Timeline:** Based on the 1st verification results, the 2nd prototyping started on August 15, 2023; production of the 2nd prototype was completed on August 25, 2023.
- **Achievements:** A total of 18 quality issues were identified during production, including 10 design & development issues, 3 process issues, 5 incoming material issues. All issues were improved and implemented during actual production.
- **Issue Identification Table:**

| 单兵车内部质量问题点清单 |      |                |      |                |                                                                                     |                                                       |               |                 |                                            |     |                  |     |
|--------------|------|----------------|------|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|---------------|-----------------|--------------------------------------------|-----|------------------|-----|
| 序号           | 区域   | 零件号            | 日期   | 问题点            | 图片                                                                                  | 原因                                                    | 问题归属          | 责任人             | 临时措施                                       | 状态  | 长期措施             | 状态  |
| 1            | 车架   | RJ2210-200-000 | 8.25 | 焊接后产生的焊脚焊点过大过高 |  | 1. 焊接工艺工装未确定。                                         | 生产过程问题        | 田祥 王毅 彭伟        | 1. 管梁打穿，在管梁底部向上焊接。<br>2. 加长螺栓<br>3. 增加焊接工装 | 已实施 | 两临时措施            | 已实施 |
| 2            | 外饰总成 | RJ2302-600-000 | 8.25 | 控制器装配后与电池仓干涉   |  | 1. 车身支架未定位。<br>2. 未料检验电池包及其上面盖不合格。<br>3. 折弯角度无工装，无管控。 | 生产过程/外购件质量/设计 | 田祥 王毅 彭伟 王翔 吕士杰 | 1. 安装板重新加工孔位                               | 已实施 | 数模上修改控制器位置       | 已实施 |
| 3            | 轮边总成 | RJ2302-301-100 | 8.25 | 前轮晃动量过大        |  | 1. 摇臂轴在一定耐久过程后无法紧固。                                   | 设计问题          | 何坤              | 无                                          | 已实施 | 1. 更改摇臂轴，增加平键和锥面 | 已实施 |

- **Test Period:** September 1, 2023 – October 1, 2023
- **Test Road Conditions:** Unpaved road conditions (professional racing venue)
- **Test Mileage:** 2,000 km
- **Test Results:** No obvious quality issues, but optimization opportunities remain:
  1. Braking Force: Uneven caliper braking force, causing vehicle body to shift right during emergency braking.
  2. Power Performance: Uneven power output.
  3. Suspension & Chassis: Slight sway in rear-wheel steering.
  4. Steering System: Steering structure is in good condition; it is recommended to increase steering angle.
- **Follow-up Actions:** all above issues were rectified in this phase. The 3rd phase verification was conducted on October 12, 2023.



**2<sup>nd</sup> Prototype Status**



**Partial Issue Photos**



**Road Test Photo**

- **Video:** 1.4 Endurance Testing Phase 2.mp4

### 2.3 Project Design Verification – 3rd Prototype

- **Prototyping Timeline:** based on results of the first two verifications, the 3rd prototyping verification started on October 12, 2023; production of the 3rd prototype was completed on November 10, 2023.
- **Achievements:** a total of 7 quality issues were identified during production, including 2 design issues, 1 process issue, 4 incoming material issues. All issues were improved and implemented during actual production.
- **Issue Identification Table:**

| 单兵车内部质量问题点清单 |      |                |      |           |                                                                                     |                    |        |            |      |     |                              |                   |
|--------------|------|----------------|------|-----------|-------------------------------------------------------------------------------------|--------------------|--------|------------|------|-----|------------------------------|-------------------|
| 序号           | 区域   | 零件号            | 日期   | 问题点       | 图片                                                                                  | 原因                 | 问题归属   | 责任人        | 临时措施 | 状态  | 长期措施                         | 状态                |
| 1            | 前悬架  | RJ2210-303-000 | 11.7 | 摆臂点焊后脆焊   |  | 1: 脆焊<br>2: 缺少工艺控制 | 生产过程问题 | 王毅林<br>田祥  | 补焊   | 已完成 | 后续增加工艺控制                     | 已实施               |
| 2            | 轮边总成 | RJ2302-301-100 | 11.7 | 电机断裂      |  | 电机外壳设计强度不足         | 供应商问题  | 王志翔<br>罗剑兵 | 无    | /   | 更换外壳材质或结构                    | 已实施待验证            |
| 3            | 转向   | RJ2302-302-300 | 11.7 | 轮胎右转过死点干涉 |  | 设计几何问题             | 设计问题   | 刘文浩        | 无    | /   | 1. 调整几何分析, 增加限位<br>2. 更改转向键点 | 目前使用八字拉杆试跑, 未发生问题 |

- **Test Period:** November 22, 2023 – January 8, 2024
- **Test Road Conditions:** Unpaved road conditions (professional racing venue) + multi-terrain testing
- **Test Mileage:** 3,000 km
- **Test Results:** No obvious quality issues, but optimization opportunities remain:
  1. Braking Force: Good braking performance, no abnormalities.
  2. Power Performance: Rear-wheel power output can be appropriately increased.
  3. Suspension & Chassis: Good chassis performance, no abnormalities.
  4. Steering System: Understeering still exists; further improvement is needed by increasing steering angle.
- **Follow-up Actions:** all above issues were rectified in this phase. The pre-production phase verification was conducted on February 10, 2024.



**3<sup>rd</sup> Prototype Status**



**Partial Issue Photos**



**Partial Road Test Photos**

- **Video:** 1.4 Endurance Testing Phase 3\_climbing.mp4  
1.4 Endurance Testing Phase 3\_stage.mp4

### 2.4 Phase 4: Project Design Verification – Pre Production Testing (PP Phase)

- **Prototyping Timeline:** based on previous verification results, the 4th phase (pre-production) verification started on February 10, 2024 (Note: Corrected from original "2023" for timeline consistency); production of the 4th prototype was completed on March 11, 2024.
- **Achievements:** Only 1 quality issue was identified during production, which comes from incoming material defect and has been improved).

#### Issue Identification Table:

|   |  |     |             |             |         |     |        |     |         |     |
|---|--|-----|-------------|-------------|---------|-----|--------|-----|---------|-----|
| 1 |  | 3.4 | 电机螺母来料规格不一致 | 供应商采购的规格不标准 | 供应商来料问题 | 王志翔 | 更改装配工装 | 已实施 | 供应商导入检具 | 已实施 |
|---|--|-----|-------------|-------------|---------|-----|--------|-----|---------|-----|

- **Test Period:** March 25, 2024 – July 1, 2024
- **Test Road Conditions:** Unpaved road conditions (professional racing venue) + multi-terrain testing
- **Test Mileage:** 6,000 km
- **Test Results:** The vehicle operated well with no obvious abnormalities.
- **Follow-up Actions:** keep testing on new features and follow-up feedback from dealers.
- **Video:** 1.4 Endurance Testing Phase 4\_water proofing.mp4  
1.4 Endurance Testing Phase 4\_desert.mp4  
1.4 Endurance Testing Phase 4\_snowy.mp4



**Vehicle Status of Pre-  
Production Phase**



**Testing Conditions  
(Water, Desert, Snowy)**

### 3. Endurance Testing Summary

After 4 rounds of prototyping verification and 13,000 km of on-road testing across multiple terrains, the vehicle currently performs well with no pending rectifications. Internal design and process plans have been finalized, and the vehicle can officially enter the mass production phase.